

Two Systems, One Map

84-355 Democracy's Data | What census geography nests, and what doesn't

Jonathan Cervas

August 10, 2026

Everything here runs as given. Your job is to read what comes out and say what it means.

Read `census-geography-brief.Rmd` first. This lab runs the same numbered steps with the code exposed. The brief tells you what the question is and why it is hard; this shows you where each number came from.

Where did that number come from?

"The data is by ZIP code."

Half the joins you will ever attempt fail, and they nearly all fail for the same reason. Somebody had data about one kind of area and needed it for a different kind of area, and assumed the two fit together.

They usually don't. American geography is two systems laid on top of each other. One nests perfectly and the other cuts across it — and knowing which is which is the difference between an analysis and a mess.

This is the lab that everything else in the course has been quietly assuming.

Steps 1-3 — One identifier, the source, and how many of each

```
lv <- read.csv("data/levels.csv", stringsAsFactors = FALSE)
lv[, c("geography", "count", "median_sq_mi")]
```

```
#>           geography count median_sq_mi
#> 1           county  3222      604.48
#> 2      census tract 85396        1.74
#> 3             place 32333        1.73
#> 4  ZCTA (ZIP approximation) 33791      34.89
#> 5      county subdivision 36421      35.59
#> 6 elementary school district  1952      33.47
#> 7 unified school district 10893     120.15
```

Seven kinds of area, from the Census Bureau's own gazetteer. **3,222 counties. 85,396 tracts. 33,791 ZCTAs. 32,333 places.**

The sizes tell you these are not variations on one idea. A median county is **604 square miles**; a median tract is **1.74**. They are different tools.

Step 4 — Test the claim the identifier makes

The identifier makes a testable claim. If an 11-digit tract GEOID really is $\text{state}(2) + \text{county}(3) + \text{tract}(6)$, then slicing off the first five digits should give a county that exists — for every tract in the country, with no exceptions. **Predict how many of the 85,396 will fail.**

Here is the thing nobody tells you, and it is the most useful fact in the course.

```
lv[, c("geography", "id_length", "id_encodes", "nests_into")]
```

```
#>           geography id_length           id_encodes
#> 1             county         5             state(2) + county(3)
#> 2      census tract        11      state(2) + county(3) + tract(6)
#> 3              place         7             state(2) + place(5)
#> 4    ZCTA (ZIP approximation)     5      five digits, nothing else
#> 5      county subdivision        10 state(2) + county(3) + subdivision(5)
#> 6 elementary school district     7             state(2) + district(5)
#> 7    unified school district     7             state(2) + district(5)
#> nests_into
#> 1      state
#> 2      county
#> 3 state only
#> 4    NOTHING
#> 5      county
#> 6 state only
#> 7 state only
```

A census GEOID is not an arbitrary key. It is an address in a hierarchy.

The 15-digit block ID mapped below reads:

```
13   153   021123   3   013
|    |    |        |   |
|    |    |        |   +- block within that block group
|    |    |        +----- block group within that tract
|    |    +----- tract within that county
|    +----- county (153 = Houston)
+----- state (13 = Georgia)
```

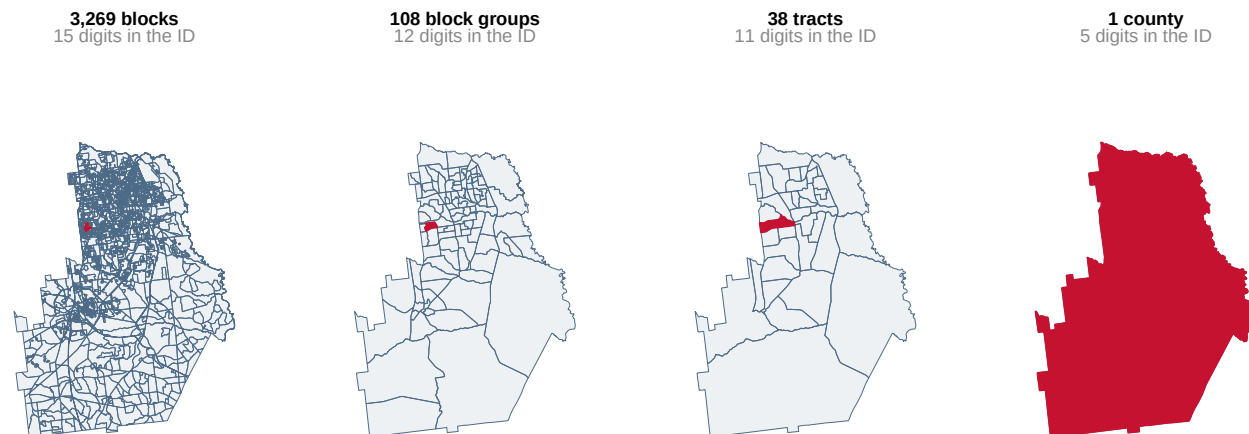
So the first five digits of any tract ID **are** a county ID. You do not need a lookup table; the join is already in the number.

```
n <- read.csv("data/nesting.csv", stringsAsFactors = FALSE)
n
```

```
#>           test result n_pass n_total
#> 1      tract GEOID[:5] is a real county    True  85396  85396
#> 2 county subdivision GEOID[:5] is a real county    True  36421  36421
#> 3              ZCTA file records a state    False    NA    NA
#> 4              place file records a county    False    NA    NA
```

All 85,396 tracts match a real county by prefix. Zero orphans. Same for all 36,421 county subdivisions. The nesting spine — block → block group → tract → county → state — is exact, and it is exact *by construction*, because the Census Bureau draws the small ones inside the big ones.

That is a claim about strings. Here is the same claim about ground: Houston County, Georgia drawn four times, with one block and each of the units containing it picked out in red.



3,269 blocks become 108 block groups become 38 tracts become one county, and **no boundary is drawn twice**. Every line in a later panel was already a line in the first one. That is what “nests” means when you can see it.

Steps 5-7 — The tests that could not be run, and what each ID encodes

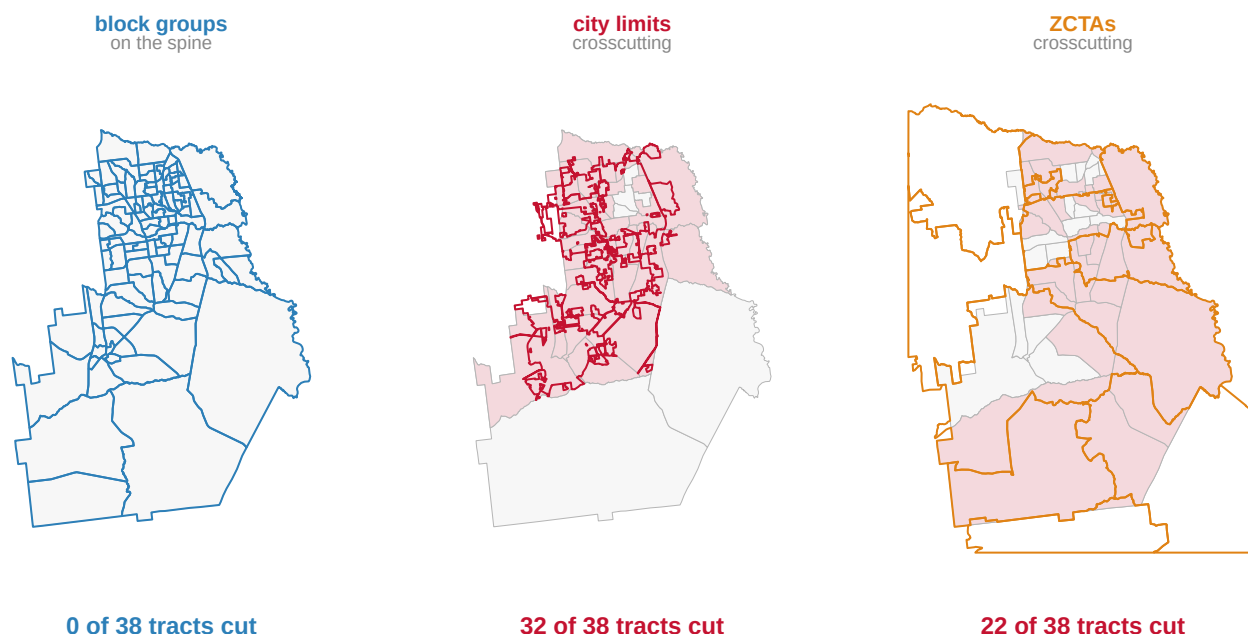
```
lv[lv$nested_into != "county" & lv$nested_into != "state",
  c("geography", "count", "id_length", "id_encodes", "nested_into")]
```

```
#>           geography count id_length      id_encodes
#> 3           place 32333         7      state(2) + place(5)
#> 4  ZCTA (ZIP approximation) 33791         5 five digits, nothing else
#> 6 elementary school district 1952         7      state(2) + district(5)
#> 7  unified school district 10893         7      state(2) + district(5)
#>  nested_into
#> 3 state only
#> 4  NOTHING
#> 6 state only
#> 7 state only
```

A place ID is seven digits: state, then place. No county. That is not an oversight — **places cross county lines**, so there is no county to record. A city is a legal entity that grew where it grew; counties were drawn separately.

Warner Robins, the city in the county we are mapping, is the example: of its 98.7 square kilometres, 91.5 are in Houston County and the rest are in the next county over. **There is no county to put in the identifier because the city is in two of them.**

Here is what that does to the tracts underneath. Same 38 tracts in all three panels; only the boundary laid over them changes. A tract is **cut** when part of it is inside a unit of the other geography and part is not.



shaded: a tract with a boundary running through it

The counts are not eyeballed off the picture. They come from the Census Bureau's own block-to-place assignment and its own ZCTA-to-tract relationship file, and they are in `splits.csv`:

```
s <- read.csv("data/splits.csv", stringsAsFactors = FALSE)
s[, c("overlay", "role", "units", "tracts_total", "tracts_split")]
```

```
#>           overlay           role units tracts_total tracts_split
#> 1      block group on the spine   108          38           0
#> 2 census place (city limits) crosscutting     4          38          32
#> 3    ZCTA (ZIP approximation) crosscutting    12          38          22
```

Zero, 32, 22. Block groups are carved out of tracts, so they cannot cross one. Cities and ZIP codes were drawn by other people for other reasons, and they cross constantly.

Same for school districts. Same, most importantly, for the one everybody uses.

Step 8 — The one everybody uses works worst

```
lv[lv$geography == "ZCTA (ZIP approximation)", ]
```

```
#>           geography count id_length           id_encodes nests_into
#> 4 ZCTA (ZIP approximation) 33791           5 five digits, nothing else   NOTHING
#>   median_sq_mi
#> 4           34.89
```

```
n[n$test == "ZCTA file records a state", ]
```

```
#>           test result n_pass n_total
#> 3 ZCTA file records a state  False     NA     NA
```

The ZCTA file has no state column, and the identifier is five digits with nothing else in it. You cannot determine what state a ZCTA is in from its ID.

That is not sloppiness. It is because **some ZIP codes are in two states**.

And it gets worse, because a ZIP code is not an area at all. **It is a set of mail delivery stops** — a route, maintained by the Postal Service, changed whenever routing changes. Some ZIP codes are a single building. Some cover no residents. They were never meant to describe where people live, and the Census Bureau’s ZCTA is a best-effort approximation of a thing that is not a shape.

```
c(zctas = lv$count[lv$geography == "ZCTA (ZIP approximation)"],
  tracts = lv$count[lv$geography == "census tract"],
  median_zcta_sq_mi = lv$median_sq_mi[lv$geography == "ZCTA (ZIP approximation)"],
  median_tract_sq_mi = lv$median_sq_mi[lv$geography == "census tract"])
```

```
#>           zctas           tracts median_zcta_sq_mi median_tract_sq_mi
#>          33791.00          85396.00           34.89           1.74
```

A median ZCTA is **34.89 square miles** — twenty times a median tract, and wildly uneven.

You gave me your ZIP code on the first day of class. It was the right thing to ask for, because it is the only geography anybody knows off the top of their head. It is also the one that will not join cleanly to anything else in this course.

Step 9 — Why this keeps happening in other labs

Every one of these was the same problem:

Where	What did not nest
Precinct returns	precincts are drawn by counties and ignore census blocks
Redlining	1937 HOLC areas against 2020 tracts
Election administration	“jurisdiction” is a township in Wisconsin, a county in Texas
Section 203	thresholds applied to jurisdictions, estimates published for others
BISG	works because blocks <i>do</i> nest — that one was the easy case

```
c(nesting_geographies = sum(lv$nested_into %in% c("state", "county")),
  crosscutting = sum(!lv$nested_into %in% c("state", "county")))
```

```
#> nesting_geographies      crosscutting
#>                   3                   4
```

When two geographies nest, the join is free and exact. When they don’t, every join is an assumption, and somebody has to decide how to split the overlap. There is no correct answer, only conventions — and the convention is usually buried in a methods appendix nobody reads.

Step 10 — What you can now say

You can say that American geography has a **spine** and a set of **crosscutting** systems; that the spine’s identifiers encode the nesting, so aggregation needs no assumptions; and that everything off the spine requires a crosswalk resting on an assumption somebody chose.

You can say which of these seven geographies are which — and you can say it because the nesting was *tested*, not read from documentation.

You cannot move data between crosscutting geographies without deciding how people are distributed within units, and no data will check that decision for you.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Build the hierarchy. Using only `levels.csv`, draw the nesting tree: which units contain which. Then mark the ones that cannot be placed in the tree at all, and say what they have in common.

Option B — How wrong is a ZIP? A median ZCTA is 34.89 square miles and a median tract 1.74 — roughly twenty tracts to a ZCTA. If someone reports a statistic “by ZIP code,” how much variation are they averaging away? Make the argument with numbers.

Option C — The school district trap. There are 1,952 elementary districts and 10,893 unified districts. A unified district covers all grades; an elementary one does not. What happens to a student living in an elementary district when they reach high school, and what does that do to anyone trying to count “students per district”?

Option D — Read the middle panel. Look at the three maps again. The city limits cut 32 of 38 tracts and the block groups cut none. Suppose you have median income by tract and you are asked for median income *in Warner Robins*. Say exactly what you would have to assume, and name one kind of neighbourhood where the assumption would fail badly.

Option E — Your own question. Seven geographies, 200,000-odd units, and one county drawn at four scales.

What this lab cannot tell you

- **Whether one county is typical.** The maps are Houston County, Georgia, and 32 of 38 is a fact about it, not a national rate. A denser county or an emptier one could be worse or better, and nothing here says which.
- **How to fix a broken join.** The standard answer is area-weighted or population-weighted allocation, and it is an approximation with its own errors. The lab shows you the problem; the fix is a decision, not a lookup.
- **Which ZIP codes cross state lines.** We can see that 5 of this county’s 12 ZCTAs cross a *county* line; the state-line cases need the same check run nationally.
- **Almost anything about people.** The county maps carry block population, but the seven national geographies are counted and measured only as containers.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- The tract ID contains the county ID. What else in this course would have been easier if every identifier worked that way?
 - A ZIP code is a mail route, not a place. Why is so much data published by ZIP anyway?
 - Which of the joins we have done this term do you now trust less?
-

Where the data came from

U.S. Census Bureau Gazetteer Files, 2024 — counties, tracts, places, ZCTAs, county subdivisions, and elementary and unified school districts. https://www2.census.gov/geo/docs/maps-data/data/gazetteer/2024_Gazetteer/

U.S. Census Bureau TIGER/Line Shapefiles, 2020 — every boundary on the maps above: the 3,269 blocks, 108 block groups, 38 tracts, 4 places and 12 ZCTAs of Houston County, Georgia. <https://www2.census.gov/geo/tiger/TIGER2020/>

U.S. Census Bureau 2020 Block Assignment Files and ZCTA relationship files — which block is in which municipality, and how ZCTA land divides among tracts. The cut counts come from these, not from our own overlay.

Everything is public and needs no key. The builds are `data/build-data.py` (the national counts) and `data/build-maps.R` (the geometry and the cut counts); the nesting tests in `nesting.csv` were run against the full files, not asserted.